RAG Assignment Report: Baseline vs. Re-Ranking

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This report answers the four areas of section 4 of the assignment. Both pipelines search the same 527 structure-aware chunks of the GDPR (Regulation (EU) 2016/679, 88 pages). They use `all-MiniLM-L6-v2` as the Bi-Encoder, `ms-marco-MiniLM-L-6-v2` as the Cross-Encoder, and `gemini-3.5-flash` (temperature 0, identical prompt) to generate the answers.

Code: `rag_rerank.ipynb`. GitHub link: https://github.com/aayush18602/rag-reranking-gdpr

0. Introduction to the dataset and how the project works

Dataset. The dataset is a single document: the EU General Data Protection Regulation (GDPR), Regulation (EU) 2016/679. It is the EU law that sets the rules for how organisations collect, store and use people's personal data, and it has applied since 25 May 2018.

I used the official 88-page English PDF from the EU Publications Office. It has two parts: 173 recitals, which explain the reasons behind the law, and 99 articles in 11 chapters, which are the binding rules.

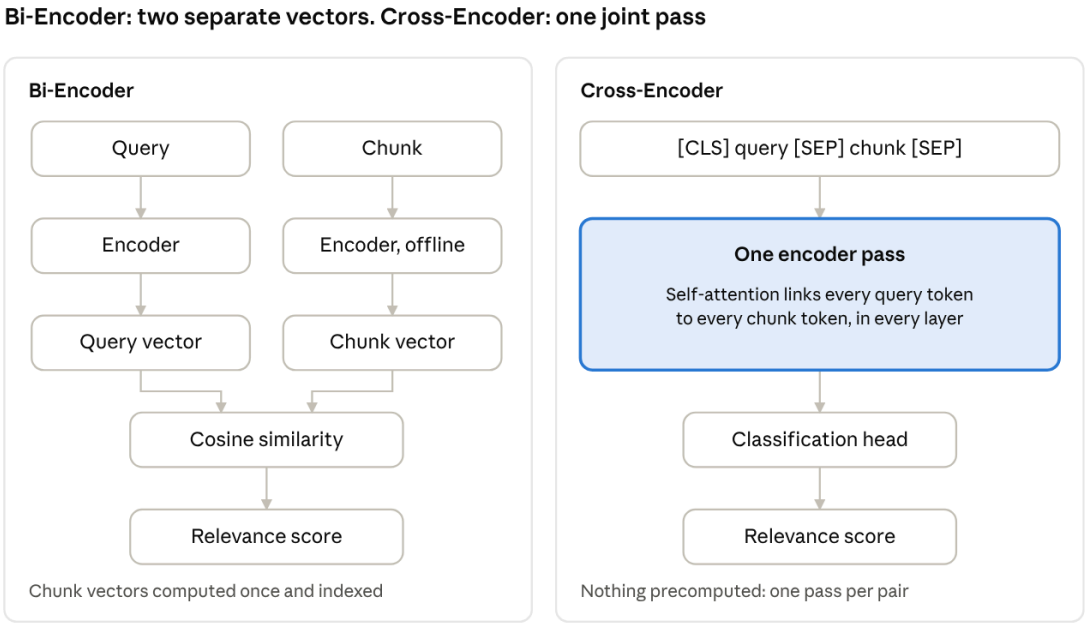
It suits this assignment for two reasons. First, it often states a general rule and then qualifies it elsewhere ("shall not apply", "shall be prohibited", "without prejudice to"). Second, it uses legal terms such as "erasure" or "dismissed" where an ordinary user would say "delete" or "fired". These are exactly the cases where a Bi-Encoder tends to retrieve the wrong passage.

How the project works. The whole system is a single Jupyter notebook that runs on a laptop CPU:

- **Cleaning:** PyMuPDF reads the PDF line by line. Page headers and footnotes are removed using their position and font size, and the paragraphs are rebuilt.
- **Chunking:** the text is split along its own structure, one recital or article at a time, with paragraphs packed into chunks of up to 150 words. This gives 527 chunks, each labelled with its source, e.g. "Article 17 – Right to erasure".
- **Indexing:** all-MiniLM-L6-v2 turns each chunk into a 384-dimensional vector, stored in a FAISS index for cosine-similarity search.
- **Pipeline 1 (baseline):** the question is embedded and the Top 10 chunks by cosine similarity are retrieved. The Top 3 are sent to Gemini, which produces Answer 1.
- **Pipeline 2 (re-ranking):** the Cross-Encoder ms-marco-MiniLM-L-6-v2 re-scores the same Top 10. The new Top 3 are sent to Gemini with the identical prompt, which produces Answer 2.
- **Evaluation:** 13 stress-test questions are run through both pipelines. For each one, the chunk that actually holds the answer is found by hand, and its rank is tracked before and after re-ranking. The two answers are then compared, and every stage is timed.

1. Architecture comparison

A Bi-Encoder encodes the query and each chunk separately, then compares the two vectors. A Cross-Encoder reads the query and the chunk together and outputs one relevance score.



Bi-Encoder vs Cross-Encoder scoring of one (query, chunk) pair

The chunk side of the Bi-Encoder runs once, offline. The Cross-Encoder has to run again for every (query, chunk) pair.

Bi-Encoder. Each text passes through the transformer alone and is pooled into one 384-dimensional vector. The chunk vectors are computed once, offline (all 527 chunks in 4.1 s). At query time only the query is encoded, and similarity is the cosine between two fixed vectors.

The weakness is that each chunk is compressed into a single vector before the query is known. Small but decisive cues get averaged into the general topic: a "not", an "except where", or which right a sentence is actually about. The Bi-Encoder is good at "what is this passage about" and weaker at "does this passage answer this question".

Cross-Encoder. The pair is fed in as one sequence, `[CLS] query [SEP] chunk [SEP]`. In every layer, self-attention lets each query token attend to each chunk token, and a classification head on top outputs a relevance score.

Because the two texts are read jointly, the model can link "fired" in the question to "shall not be dismissed" in the chunk, and see whether the chunk states the rule or only mentions the topic. The price: nothing can be precomputed, so every (query, chunk) pair needs its own full forward pass.

	Bi-Encoder	Cross-Encoder
Input	One text at a time	Query and chunk concatenated
Query–chunk interaction	None until the final cosine	Full self-attention across both, in every layer

	Bi-Encoder	Cross-Encoder
Output	A vector per text	A single relevance score per pair
Precomputable	Yes, chunk vectors are indexed once	No, it depends on the query
Work per query	1 encoder pass + a vector search	1 encoder pass per candidate chunk
Role here	Fast recall of 10 candidates	Precise ordering of those 10

2. Qualitative & rank shift analysis

Re-ranking moved the gold chunk into the Top 3 for 8 of 13 stress-test queries. In the 6 cases run through the LLM, Answer 1 was wrong or incomplete every time and Answer 2 was correct every time.

Method. For each question I first found the *gold* chunk, the chunk that actually contains the answer, by matching a distinctive phrase from the legal text. I then recorded its rank in the baseline ordering (over the whole corpus) and in the re-ranked Top 10. Questions are phrased in everyday words ("fired", "delete", "small violation") while the law uses legal terms ("dismissed", "erasure", "minor infringement"). That vocabulary gap is what the stress tests probe.

2.1 Rank changes

Failure type	Query	Gold chunk	Rank (baseline → re-ranked)	Outcome
Keyword trap / distractor	Can a supervisory authority just give a warning instead of a fine for a small violation?	Recital 148 (#220)	#4 → #1	Fixed
Keyword trap / synonym	Can I ask a company to delete my data?	Article 17(1) (#311)	#4 → #2	Fixed
Exception / corner case	Can a country limit my data protection rights for reasons of national security?	Article 23 (#324)	#9 → #2	Fixed
Exception / corner case	Can the right to get a copy of my data be limited to protect other people?	Article 15(4) (#309)	#5 → #1	Fixed
Exception / corner case	Can I object to direct marketing even if the company has a legitimate interest?	Article 21(2) (#319)	#5 → #1	Fixed
Negation & constraints	Can the data protection officer be fired for doing their job?	Article 38(3) (#368)	#8 → #3	Fixed

Failure type	Query	Gold chunk	Rank (baseline → re-ranked)	Outcome
Buried chunk	Is pseudonymised data still considered personal data?	Recital 26 (#32)	#9 → #3	Fixed
Buried chunk	Can an organisation file a complaint on my behalf?	Recital 142 (#208)	#6 → #2	Fixed
Counter-example	Is a processor liable for damages in the same way as the controller?	Article 82(2) (#495)	#1 → #4	Hurt
Counter-example	When is a data protection impact assessment mandatory?	Article 35(3) (#355)	#6 → #8	Missed both times
Limitation (negation)	Which types of personal data is a company not allowed to process?	Article 9(1) (#282)	#86 → not in Top 10	Not retrieved
Limitation (negation)	Are courts supervised by data protection authorities?	Article 55(3) (#424)	#150 → not in Top 10	Not retrieved
Limitation (keyword trap)	How much can a company be fined for breaking the basic principles for processing?	Article 83(5) (#501)	#11 → not in Top 10	Not retrieved

2.2 Answer 1 vs Answer 2

Query (gold rank)	Answer 1, baseline	Answer 2, re-ranked	What the baseline got wrong
National security (Art. 23, #9 → #2)	"There is no information about whether a country can limit data protection rights for reasons of national security."	Yes. A legislative measure may restrict the rights in Articles 12–22 and 34, if it respects the essence of fundamental rights and is necessary and proportionate.	**False "not covered".** Its Top 3 were Recitals 104, 114 and 11, which mention national security in the context of transfers to third countries and the Regulation's general aims.
DPO fired (Art. 38(3), #8 → #3)	"There is no information about whether a data protection officer can be fired."	No. The DPO "shall not be dismissed or penalised … for performing his tasks".	**False "not covered".** All 3 baseline chunks were about the DPO (other paragraphs of Arts. 37–39). Right topic, wrong paragraph.

Query (gold rank)	Answer 1, baseline	Answer 2, re-ranked	What the baseline got wrong
Warning instead of fine (Recital 148, #4 → #1)	"It is not explicitly stated"; quotes Recital 150's "giving a warning does not affect … other powers".	Yes. "In a case of a minor infringement … a reprimand may be issued instead of a fine."	**Keyword trap.** Recital 150 contains the literal word "warning" and won; the rule itself uses "minor infringement" and "reprimand".
Copy of my data limited (Art. 15(4), #5 → #1)	Yes, citing Recital 68: the right to receive data is "without prejudice to the rights and freedoms of other data subjects".	Yes. "The right to obtain a copy … shall not adversely affect the rights and freedoms of others" (Art. 15(4)).	**Wrong legal basis.** Recital 68 is about data *portability*, a different right. Right conclusion, misattributed rule.
Delete my data (Art. 17(1), #4 → #2)	Yes, but lists only the legal-obligation and child-data grounds for erasure.	Yes, listing the main grounds: data no longer necessary, consent withdrawn, objection, unlawful processing.	**Omitted clauses.** The baseline had the second half of Article 17 but not paragraph 1, so the four main grounds were missing.
Pseudonymised data (Recital 26, #9 → #3)	"The passages do not explicitly state whether pseudonymised data is still considered personal data."	Yes. Pseudonymised data "should be considered to be information on an identifiable natural person".	**Non-answer.** It had the definition (Art. 4) and the anonymous-data half of Recital 26, but not the sentence that answers the question.

The baseline never invented a rule outright. Because the prompt says "answer only from the context", its failures were false "not covered" answers (3 of 6), a hedge, a wrong legal source and missing clauses. Without that instruction, the same missing chunks would likely have produced hallucinated answers from general knowledge.

2.3 Where re-ranking does not help

- **It can hurt.** For processor liability, the baseline ranked the exact clause first (Art. 82(2): a processor is liable "only where it has not complied with obligations … specifically directed to processors"). The Cross-Encoder pushed it to #4 (score 7.20) behind three general compensation passages (8.01–8.48). The model is trained on MS MARCO web-search queries, not legal text, and when scores are this close the order can flip. The LLM answers for this case were not generated, because the free-tier daily quota ran out.
- **It cannot recover what was never retrieved.** The Cross-Encoder only re-orders the Top 10. Article 9(1), the list of prohibited data types, ranked #86 for "not allowed to process": the negated question matched recitals about lawful processing in general. Article 83(5) ranked #11, just outside the cut-off, so a Top 20 would have caught it. A larger candidate pool, or a keyword (BM25) retriever alongside the dense one, would address this.

3. Latency profiling

Re-ranking makes retrieval about 14× slower (97.6 ms vs 6.8 ms), but that is only about 91 ms on top of an LLM call of about 5.7 s, roughly 1.6% of end-to-end time.

Retrieval stages were timed with `time.perf_counter()` on a laptop CPU, after one warm-up call: 13 queries × 5 repeats = 65 runs each. LLM time is the network round-trip of each Gemini call made in the notebook.

Stage	Mean (ms)	Std (ms)	Runs	vs baseline
Pipeline 1: embed query + FAISS Top 10	6.8	1.0	65	1×
Cross-Encoder re-rank of the 10 candidates	90.7	4.6	65	13.3×
Pipeline 2: retrieval + re-rank	97.6	4.8	65	14.3×
Cross-Encoder over all 527 chunks (no Bi-Encoder)	3,944.6	59.3	3	578×
LLM generation, one call (`gemini-3.5-flash`)	5,724.6	2,031.8	15	839×

Indexing is a one-off cost: embedding all 527 chunks took 4.1 s. Re-ranking cost grows linearly with the number of candidates (about 9 ms per pair here), so re-ranking a Top 50 instead of a Top 10 would still cost under 0.5 s.

4. Scaling question

We don't run the Cross-Encoder over the whole collection because it needs a full transformer pass for every chunk on every query, and none of that work can be done in advance. On this small 527-chunk document it already took 3.9 s per query, 578× the baseline.

- **Cost grows with the collection.** At about 7.5 ms per (query, chunk) pair on this CPU, a collection of 1 million chunks would take about 2 hours per query. Every new query pays the full cost again.
- **Nothing can be precomputed or indexed.** The Cross-Encoder's score depends on the query and the chunk together, so it cannot be cached. Bi-Encoder chunk vectors are computed once, and a vector index (FAISS, or an approximate index at larger scale) searches millions of them in milliseconds.
- **Two stages give the best of both.** The cheap Bi-Encoder narrows the collection to K candidates with high recall. The expensive Cross-Encoder then orders only those K precisely. The cost per query becomes 1 query encoding, 1 vector search and K pair passes: fixed, whatever the size of the collection.

The trade-off is that stage 1 caps recall. A chunk the Bi-Encoder ranks below K is never seen by the re-ranker (Article 9(1) at #86 in section 2.3). K is therefore a dial between recall and latency.